

Lab 4: RLC Circuits - Part 2

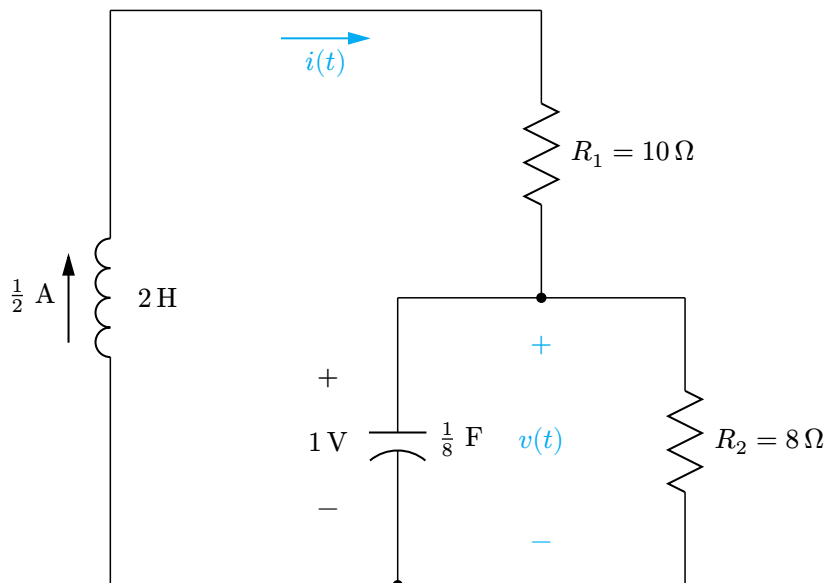
ECE2200L Section 04, Fall 2026

Matthew Ponciano, Eli Pantoja

September 16, 2026 - September 21, 2026

No 1.

Step 1. Given the circuit shown, calculate the expression for $v(t)$ for $t \geq 0$



notice

$$v(t) = v_C(t)$$

$$v(0^+) = 1 \text{ V}$$

$$i_L(0^+) = \frac{1}{2} \text{ A}$$

and using KCL:

$$\begin{aligned} i(t) &= i_C + i_{R_2} \\ &= C \frac{dv}{dt} + \frac{1}{R_2} v \end{aligned}$$

$$i(t) = \frac{1}{8} v' + \frac{1}{8} v$$

$$i'(t) = \frac{1}{8} v'' + \frac{1}{8} v'$$

now use KVL:

$$0 = v_L + R_1 i + v$$

$$0 = L \frac{di}{dt} + R_1 i + v$$

$$0 = 2 \left(\frac{1}{8} v'' + \frac{1}{8} v' \right) + 10 \left(\frac{1}{8} v' + \frac{1}{8} v \right) + v$$

$$0 = \frac{1}{4} v'' + \frac{1}{4} v' + \frac{5}{4} v' + \frac{5}{4} v + v$$

$$0 = v'' + 6v' + 9v$$

writing the characteristic equation and solving:

$$s^2 + 6s + 9 = 0$$

$$(s + 3)^2 = 0$$

this is a critically damped circuit with a double root of -3 :

$$v(t) = (A + Bt)e^{-3t}$$

now recall $v(0) = 1 \text{ V}$. substituting into the solution for $v(t)$ yields

$$A = 1$$

to find B , use the KCL equation at $t = 0$:

$$i(0) = C \frac{dv(0)}{dt} + \frac{1}{R_2} v(0)$$

$$\frac{1}{2} = \frac{1}{8} v'(0) + \frac{1}{8}$$

$$v'(0) = 3 \text{ V/s}$$

now differentiating the solution,

$$v'(0) = B - 3A$$

$$B = 6$$

thus,

$$\boxed{v(t) = (1 + 6t)e^{-3t} \text{ V}} \text{ for } t \geq 0$$

Step 2. Simulate the circuit to confirm your result in Step 1.

[View simulation code](#)

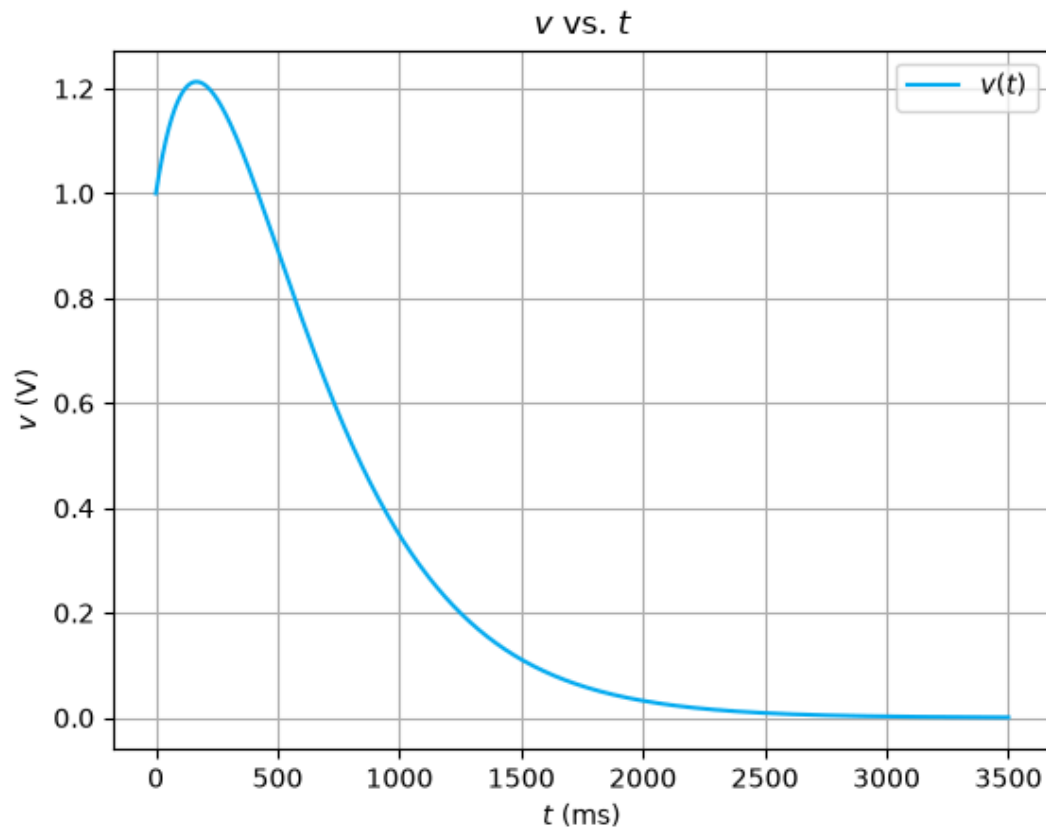
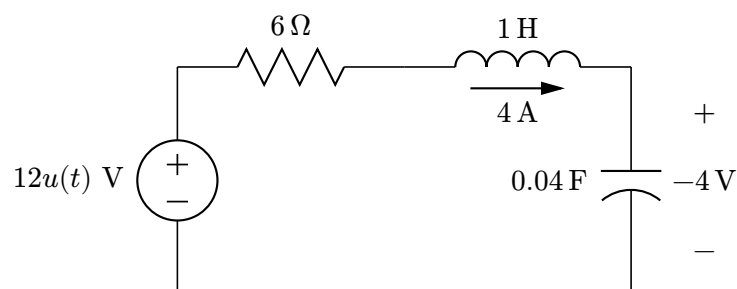


Figure 1: Simulated voltage $v(t)$, confirming the critically damped response obtained analytically. The voltage begins at 1 V, rises slightly due to the initial inductor current, and then decays smoothly to 0 V without oscillation.

No 2.

Step 1. Given the circuit shown, calculate the expression for $v_C(t)$ for $t \geq 0$



for a series RLC configuration:

$$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\alpha = 3 \text{ s}^{-1}, \quad \omega_0 = 5 \text{ rad/s}$$

since $\alpha < \omega_0$, the system is underdamped, with

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 4$$

this is a step response with $v_C(\infty) = 12 \text{ V}$, so

$$v_C(t) = 12 + e^{-3t}[A \cos(4t) + B \sin(4t)]$$

notice

$$v_C(0^+) = v_C(0^-) = -4 \text{ V}$$

so at $t = 0$,

$$-4 = 12 + A$$

$$A = -16$$

now to obtain B , notice

$$i(0^+) = 4 \text{ A}$$

then

$$i = C \frac{dv_C}{dt}$$

$$\frac{dv_C(0^+)}{dt} = \frac{i(0^+)}{C}$$

$$\frac{dv_C(0^+)}{dt} = 100 \text{ V/s}$$

now differentiate the solution and evaluate it at $t = 0$:

$$v'(0) = -\alpha A + \omega_d B$$

$$B = \frac{v'(0) + \alpha A}{\omega_d}$$

$$B = 13$$

thus,

$$\boxed{v_C(t) = 12 + e^{-3t}[-16 \cos(4t) + 13 \sin(4t)] \text{ V}} \text{ for } t \geq 0$$

Step 2. Simulate the circuit to confirm your result in Step 1.

[View simulation code](#)

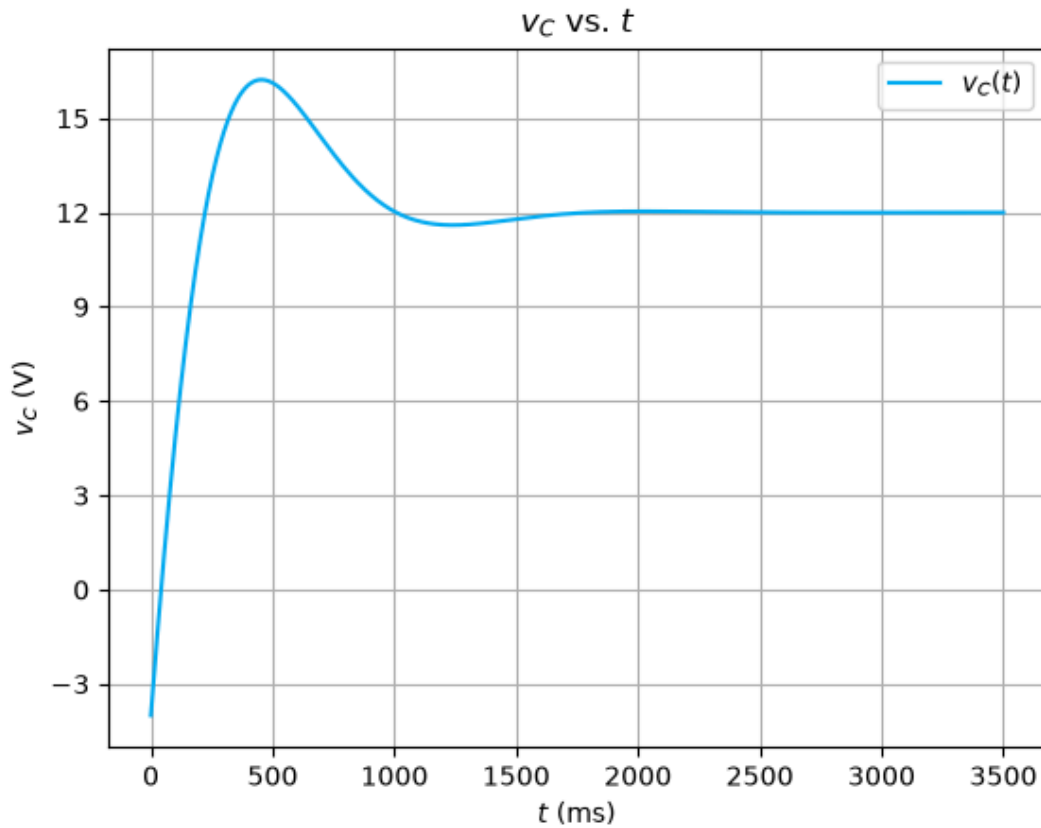
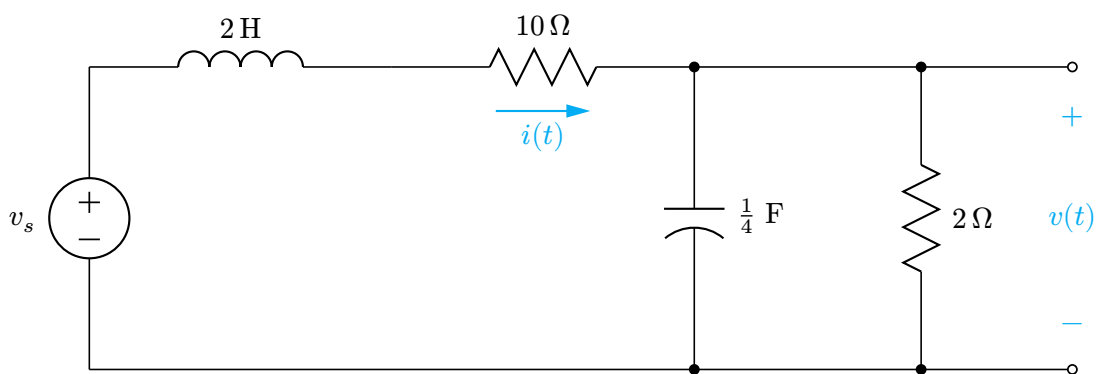


Figure 2: Simulated capacitor voltage $v_C(t)$, showing the expected underdamped step response. The capacitor voltage begins at -4 V , overshoots the 12 V steady-state value, and then settles to 12 V through decaying oscillations.

No 3.

Step 1. Given the circuit shown, calculate the expression for $v(t)$ for $t \geq 0$



where

$$v_s = \begin{cases} 12\text{ V} & \text{for } t < 0 \\ 24\text{ V} & \text{for } t > 0 \end{cases}$$

at $t = 0^-$, the circuit is in DC steady state, so the inductor behaves as a short circuit and the capacitor as an open circuit

$$i_L(0^+) = i_L(0^-) = \frac{12}{10 + 2} = 1 \text{ A}$$

and it follows that the voltage across the capacitor is the voltage across R_2 :

$$v_C(0^+) = i_L(0^+)R_2 = 2 \text{ V}$$

now we derive the equations for $t > 0$. notice:

$$v_s = 24 \text{ V}$$

$$i = i_C + i_{R_2}$$

$$v(t) = v_C(t)$$

so,

$$i = Cv' + \frac{1}{R_2}v$$

$$i = \frac{1}{4}v' + \frac{1}{2}v$$

$$i' = \frac{1}{4}v'' + \frac{1}{2}v'$$

now do a KVL around the leftmost loop:

$$0 = 24 - Li' - R_1i - v$$

$$24 = Li' + R_1i + v$$

$$24 = 2\left(\frac{1}{4}v'' + \frac{1}{2}v'\right) + 10\left(\frac{1}{4}v' + \frac{1}{2}v\right) + v$$

$$24 = \frac{1}{2}v'' + v' + \frac{10}{4}v' + 5v + v$$

$$48 = v'' + 7v' + 12v$$

write the characteristic equation:

$$s^2 + 7s + 12 = 0$$

$$(s + 3)(s + 4) = 0$$

$$s \in \{-3, -4\}$$

two real roots indicate the solution is overdamped. to find the steady-state value of $v(t)$, set the derivatives to 0 ($v(t)$ stops changing as $t \rightarrow \infty$)

$$12v(\infty) = 48$$

$$v(\infty) = 4$$

therefore, the solution has the form

$$v(t) = 4 + Ae^{-3t} + Be^{-4t}$$

to find A and B , recall that $v(0^+) = 2 \text{ V}$. at $t = 0$,

$$4 + A + B = 2$$

$$A + B = -2$$

and $i(0^+) = 1 \text{ A}$:

$$i = \frac{1}{4}v' + \frac{1}{2}v$$

$$1 = \frac{1}{4}v'(0) + \frac{1}{2}(2)$$

$$v'(0) = 0$$

now differentiate the response and evaluate at $t = 0$:

$$-3A - 4B = 0$$

$$3A + 4B = 0$$

solving the system:

$$\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} -8 \\ 6 \end{bmatrix}$$

thus,

$$\boxed{v(t) = 4 - 8e^{-3t} + 6e^{-4t} \text{ V}} \text{ for } t \geq 0$$

Step 2. Simulate the circuit to confirm your result in Step 1.

[View simulation code](#)

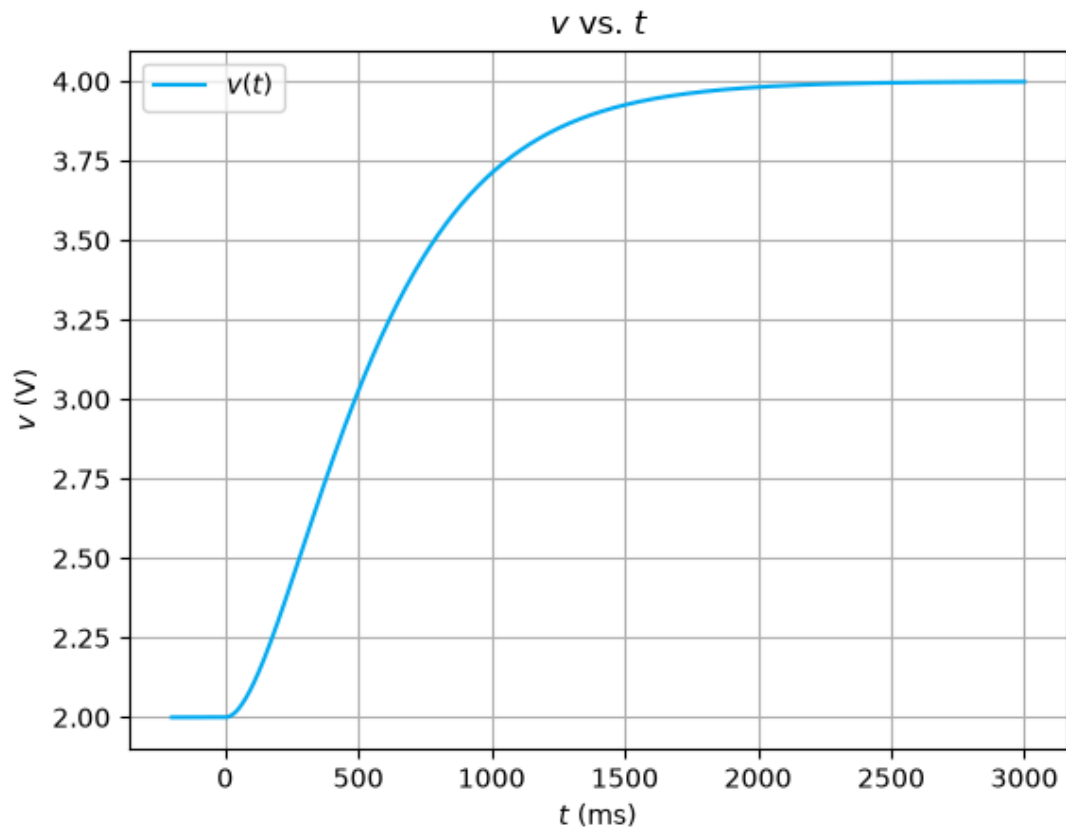


Figure 3: Simulated output voltage $v(t)$, confirming the overdamped response predicted analytically. The voltage begins at 2 V and rises monotonically toward its 4 V steady-state value without overshoot or oscillation.